

◆ R-TYPE

Addr	Mnemonic	설명
0	add x4, x2, x1	$x4 \leftarrow x2 + x1$
1	sub x5, x2, x1	$x5 \leftarrow x2 - x1$
2	and x6, x3, x0	$x6 \leftarrow x3 \& x0$
3	or x7, x3, x0	$x7 \leftarrow x3 x0$
4	sll x8, x1, x2	$x8 \leftarrow x1 \ll x2$
5	srl x9, x1, x2	$x9 \leftarrow x1 \gg x2$ (논리)
6	sra x10, x1, x2	$x10 \leftarrow x1 \ggg x2$ (산술)

◆ I-TYPE (연산/Shift)

Addr	Mnemonic	설명
7	addi x11, x1, 1	$x11 \leftarrow x1 + 1$
8	andi x12, x2, 4	$x12 \leftarrow x2 \& 4$
9	ori x13, x2, 1	$x13 \leftarrow x2 1$
10	slli x14, x1, 3	$x14 \leftarrow x1 \ll 3$
11	srli x15, x1, 2	$x15 \leftarrow x1 \gg 2$ (논리)
12	srai x16, x1, 2	$x16 \leftarrow x1 \ggg 2$ (산술)

◆ S/L-TYPE

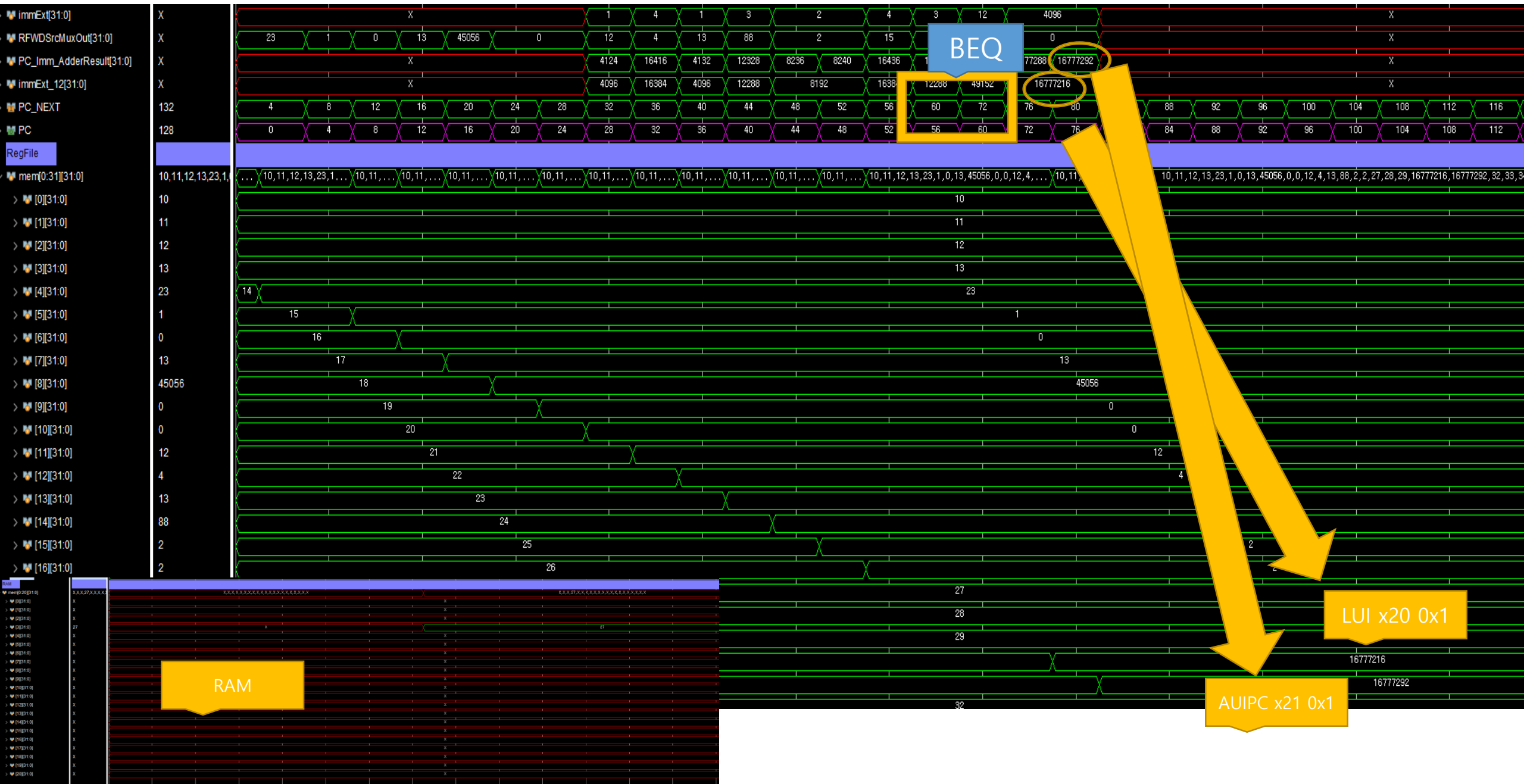
Addr	Mnemonic	설명
13	sw x17, 4(x1)	$\text{Mem}[x1+4] \leftarrow x17$
14	lw x17, 3(x1)	$x17 \leftarrow \text{Mem}[x1+3]$

◆ B-TYPE

Addr	Mnemonic	설명
15	beq x2, x2, 12	if($x2=x2$) $\text{PC} \leftarrow \text{PC}+12$

◆ J/U-TYPE

Addr	Mnemonic	설명
16	jal x18, 8	$x18 \leftarrow \text{PC}+4$; $\text{PC} \leftarrow \text{PC}+8$
17	jalr x19, x18, 0	$x19 \leftarrow \text{PC}+4$; $\text{PC} \leftarrow x18$
18	lui x20, 0x1	$x20 \leftarrow 0x1 \ll 12$ (=4096)
19	auipc x21, 0x1	$x21 \leftarrow \text{PC} + (0x1 \ll 12)$



0	0	<code>nop</code>	—	
1	4	<code>addi x1, x0, 5</code>	x1 = 5	
2	8	<code>addi x2, x0, 10</code>	x2 = 10	
3	12	<code>addi x3, x0, 7</code>	x3 = 7	
4	16	<code>addi x4, x0, 12</code>	x4 = 12	
5	20	<code>jal x10, +8</code>	x10 = PC+4 = 24, PC → 28	rom[6] 스킵
6	24	<code>addi x11, x0, 1</code>	(실행 안 됨)	스킵
7	28	<code>addi x12, x0, 2</code>	x12 = 2	
8	32	<code>addi x13, x0, 3</code>	x13 = 3	
9	36	<code>auipc x4, 0</code>	x4 = PC(=36)	<code>addr(rom[9])</code>
10	40	<code>addi x4, x4, 16</code>	x4 = 52	점프 타깃 주소 준비
11	44	<code>jalr x11, x4, 0</code>	x11 = PC+4 = 48, PC → (52 & ~1) = 52	rom[12] 스킵
12	48	<code>addi x14, x0, 15</code>	(실행 안 됨)	스킵
13	52	<code>addi x15, x0, 16</code>	x15 = 16	JALR 타깃
14	56	<code>addi x16, x0, 17</code>	x16 = 17	
15	60	<code>jal x1, +8</code>	x1 = PC+4 = 64, PC → 68	서브루틴 진입
16	64	<code>addi x17, x0, 32</code>	복귀 후 실행: x17 = 32	리턴 지점
17	68	<code>addi x20, x0, 5</code>	x20 = 5	서브루틴 내부
18	72	<code>jalr x0, x1, 0</code>	PC → x1 = 64	rom[16]으로 복귀
19	76	<code>addi x21, x0, 33</code>	x21 = 33	복귀 이후

